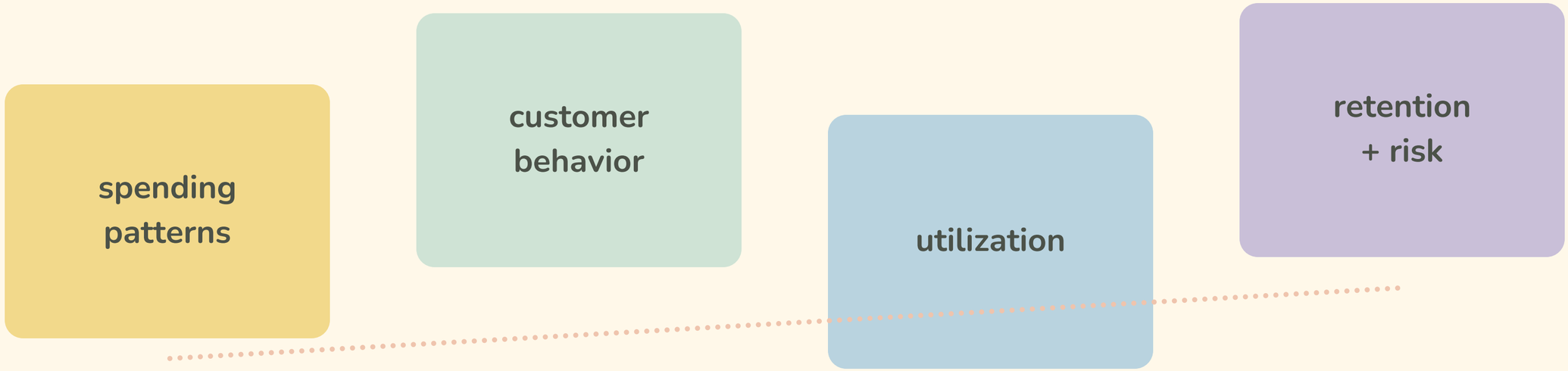
FINANCIAL ANALYSIS PROJECT

A little journey through credit card data

Presented by Kunal Malviya

What are we curious about?

Exploring spending patterns, customer behavior, utilization, retention, and risk with Power BI and DAX.



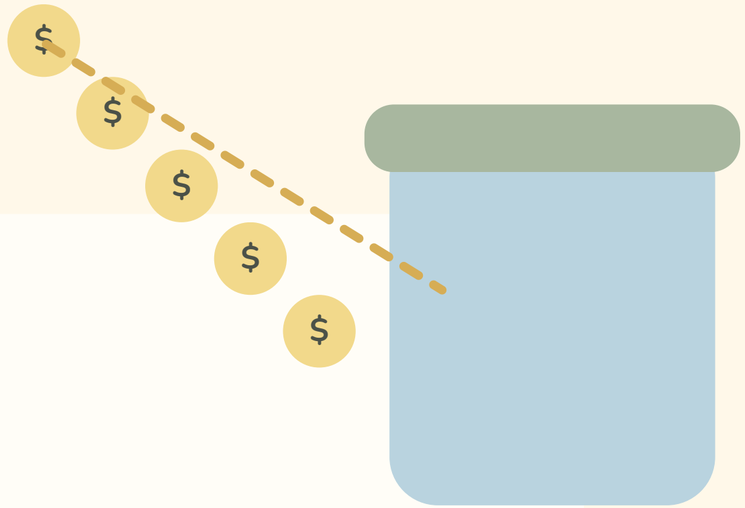
Metrics

- Running total of Credit Card transactions
- 4-Week Moving Average of Credit Limit for Each Client
- MOM% Growth and Wow% Growth on transaction Amount
- Customer Acquisition Cost (CAC) as a Ratio of transaction Amount
- Yearly Average of Avg_utilization_Ratio for All Clients
- Percentage of Interest Earned Compared to total Revolving Balance for Each Client
- Top 5 Clients by total transaction Amount
- Clients Whose Avg_utilization_Ratio Exceeds 80%
- Customer Churn Indicator
- Delinquency Rate
- Credit Risk Score
- Average customer Satisfaction Score by Credit Card Category
- Loan Approval vs Credit Limit Analysis

01 ·

The money river

Running total of credit card transactions

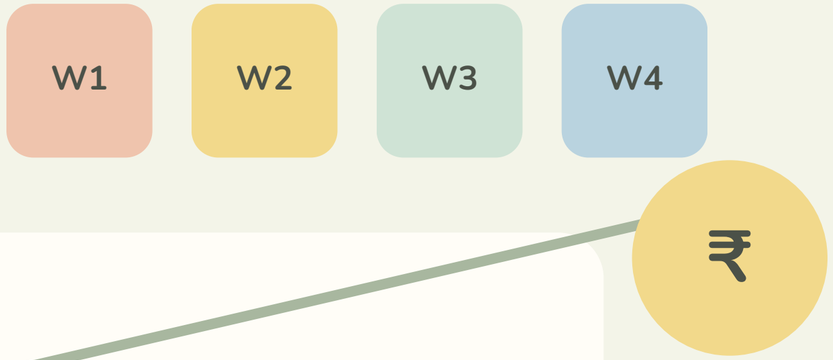



```
running_transaction =
 CALCULATE(
        SUM(credit_card[Total_Trans_Amt]),
        FILTER(
                ALL(credit_card[Week_Start_Date]),
                credit_card[Week_Start_Date] <= MAX(credit_card[Week_Start_Date])
         )
)
```

02 ·

Four-week balance check

A 4-week moving average of Credit_Limit for each client smooths short-term changes and reveals the underlying pattern.




```
MA_four_Weeks =

VAR time_period = FILTER(
            ALL(credit_card),
            credit_card[Week_Num] >= MAX(credit_card[Week_Num]) - 3
           && credit_card[Week_Num] <= MAX(credit_card[Week_Num])
           && credit_card[current_year] = MAX(credit_card[current_year])
)

VAR credit_limit_time_period = CALCULATE(SUM(credit_card[Credit_Limit]), time_period)

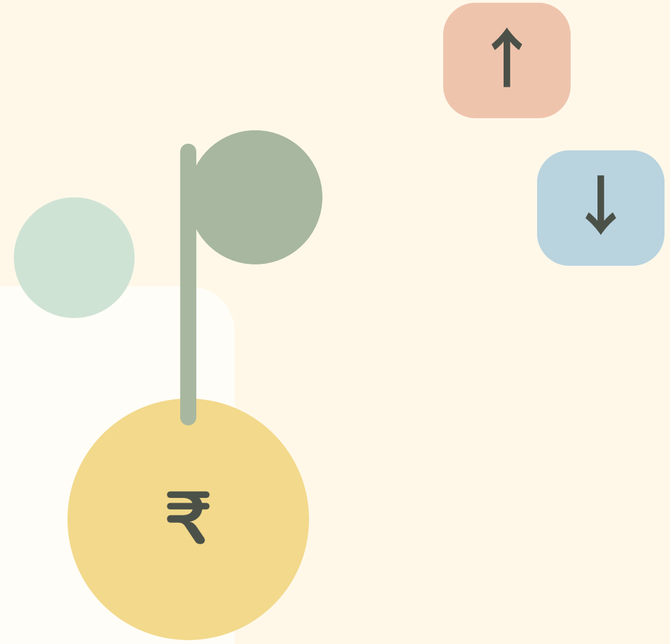
VAR distinct_weeks = CALCULATE(DISTINCTCOUNT(credit_card[Week_Num]), time_period)

RETURN DIVIDE(credit_limit_time_period, distinct_weeks, 0)
```

03 ·

Growth sprout

MOM% Growth on transaction Amount




```
mom_growth_on_transaction_amt =

VAR prev_month_transaction =
                    CALCULATE(
                            [total_transaction],
                            DATEADD('calendar'[Date], -1, MONTH)
                    )

RETURN DIVIDE(
                [total_transaction] - prev_month_transaction,
                prev_month_transaction, 0)
```

03 ·

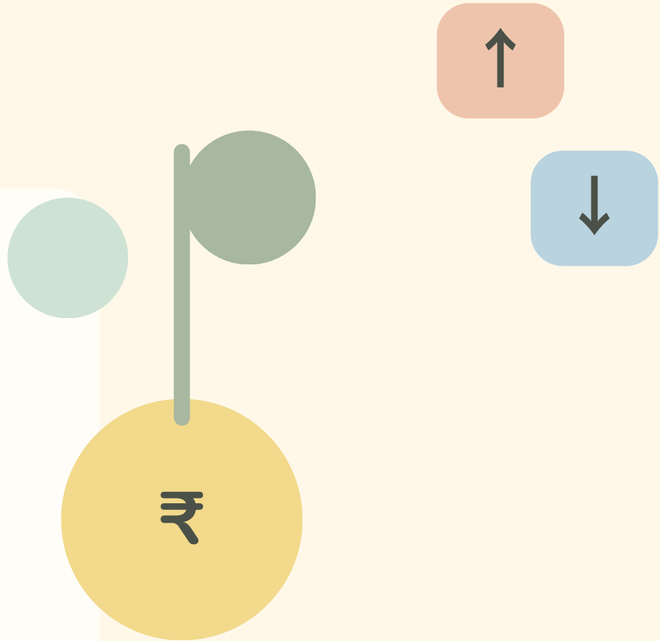
Growth sprout

WOW% Growth on transaction Amount

```
wow_transaction_growth =

VAR prev_week_transaction =
                CALCULATE(
                        [total_transaction],
                        FILTER(ALL('calendar'),
                        'calendar'[weeknum] = MAX('calendar'[weeknum]) - 1
                        && 'calendar'[year] = MAX('calendar'[year])
                )
)

RETURN DIVIDE(
        [total_transaction] - prev_week_transaction,
        prev_week_transaction, 0)
```



04 ·

Cost to welcome a customer

Customer Acquisition Cost (CAC) as a Ratio of transaction Amount

```
total_transaction = SUMX(
                        credit_card,
                        credit_card[Total_Trans_Amt])
```

```
total_cac = SUMX(
                    credit_card,
                    credit_card[Customer_Acq_Cost])
```

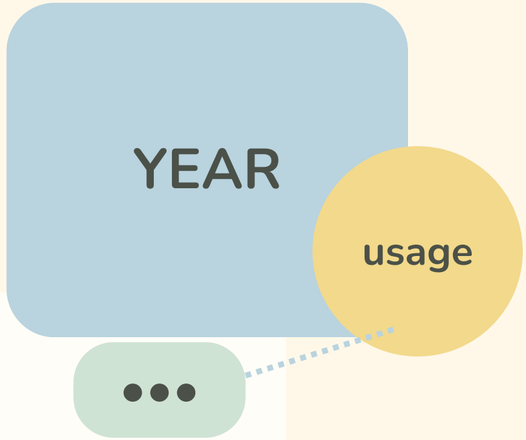
```
cac_ratio_of_transaction_amt =
                        DIVIDE([total_cac],
                        [total_transaction],
                        0)
```



05 ·

The yearly usage garden

Yearly Average of Avg_utilization_Ratio for All Clients



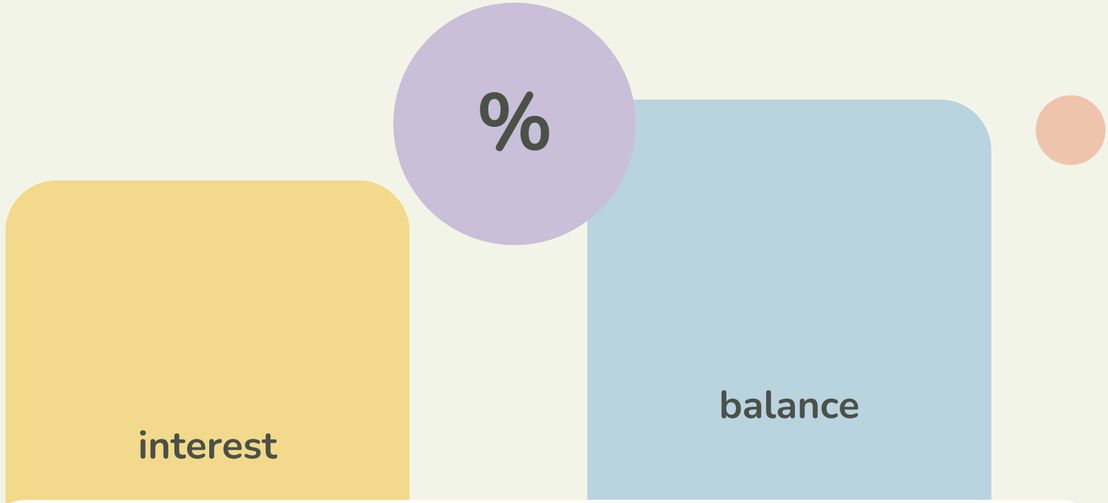
yearly_average_utilization =

AVERAGE(credit_card[Avg_Utilization_Ratio])

06 ·

Interest meets balance

Percentage of Interest Earned Compared to Total Revolving Balance for Each Client




```
total_interest_earned =
                SUMX(credit_card,
                    credit_card[Interest_Earned])
```

```
total_revolving_bal =
                SUMX(credit_card,
                    credit_card[Total_Revolving_Bal])
```

```
interest_earned_to_revolving_bal =
                        DIVIDE([total_interest_earned],
                            [total_revolving_bal],
                            0
                        )
```

07 ·

The golden five

Top 5 Clients by total transaction Amount

1 client

2 client

3 client

4 client

5 client

```
top_5_clients_by_transaction_amt =

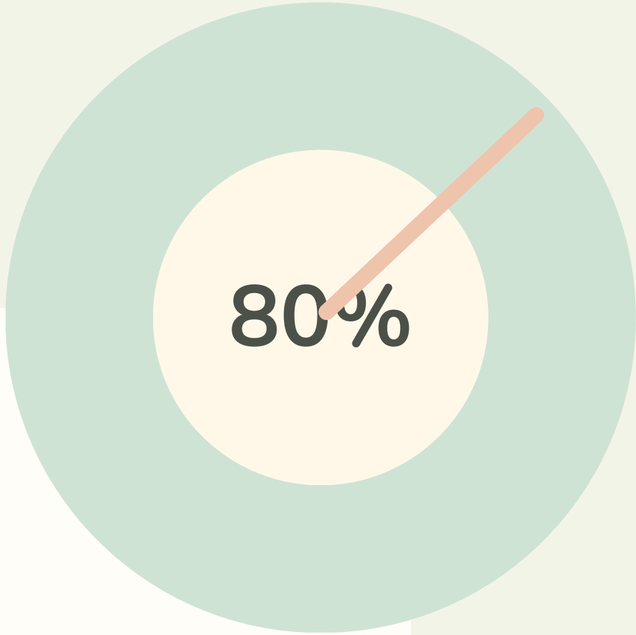
VAR ranking = RANKX(
                    ALLSELECTED(credit_card[Client_Num]),
                                [total_transaction], ,
                                DESC, Dense
                    )

RETURN IF(
                     ranking <= 5,
                    [total_transaction]
                    )
```

08 ·

The 80% line

Clients Whose Avg_utilization_Ratio Exceeds 80%




```
exceed_80%_utilization =

VAR percent_avg_utilization_ratio =
                                    SUMX(
                                        credit_card,
                                        credit_card[Avg_Utilization_Ratio]
                                    ) * 100

RETURN IF(
                [percent_avg_utilization_ratio] > 80.00,
                [percent_avg_utilization_ratio])
```

09 ·

The quiet wallet

Customer Churn Indicator: Create a KPI that flags clients who have not made any transactions (Total_Trans_Amt = 0) in the last 6 months.




```
churn_indicator =

VAR max_date = MAX(credit_card[Week_Start_Date])

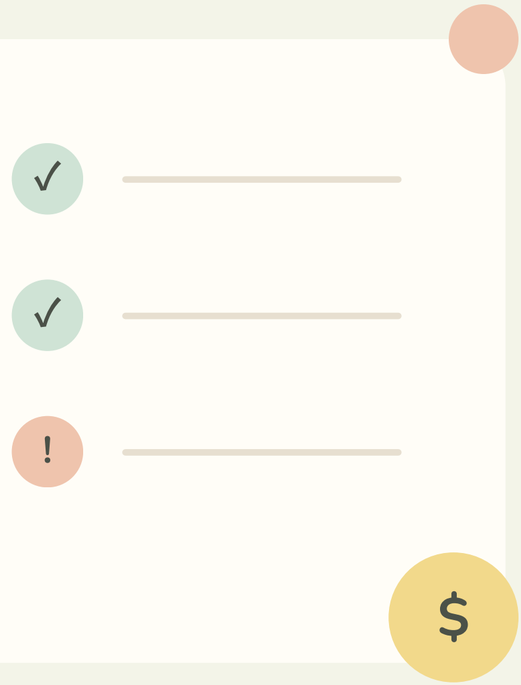
VAR max_date_of_dataset =
                    CALCULATE(MAX(credit_card[Week_Start_Date]), ALL(credit_card))

VAR month_transation =
                    CALCULATE(
                        [total_transaction],
                        DATESINPERIOD(credit_card[Week_Start_Date],
                        max_date_of_dataset, -6, MONTH))

RETURN IF(month_transation = 0 , "churned", "not_churned")
```

A little delinquency check

Delinquency Rate: Calculate the percentage of clients with Delinquent_Acc > 0.




```
delinquency_rate =

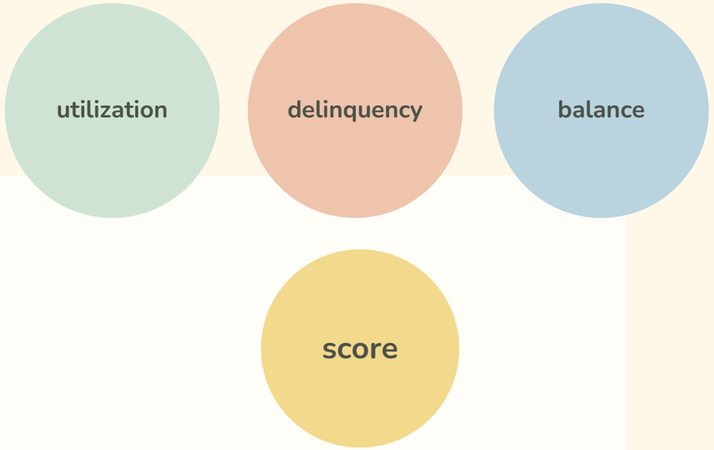
VAR deliquent_client =
                COUNTROWS(
                        FILTER(credit_card, credit_card[Delinquent_Acc] >0))

VAR total_client = DISTINCTCOUNT(credit_card[Client_Num])

RETURN DIVIDE(deliquent_client, total_client, 0)
```

The risk recipe

Credit Risk Score: Create a score for each client based on their Avg_Utilization_Ratio, Delinquent_Acc, and Total_Revolving_Bal.




```
credit_risk_score =

VAR utilization = SELECTEDVALUE(credit_card[Avg_Utilization_Ratio])

VAR deliquent = SELECTEDVALUE(credit_card[Delinquent_Acc])

VAR revolving_bal = SELECTEDVALUE(credit_card[Total_Revolving_Bal])

VAR utilization_points = SWITCH(TRUE(), utilization < 0.30, 0, utilization < 0.70, 1, 2)

VAR deliquent_points = IF(deliquent = 1, 2, 0)

VAR revolving_points = IF(revolving_bal > 0, 1, 0)

RETURN utilization_points + deliquent_points + revolving_points
```

Loan yes or no?

Loan Approval vs Credit Limit: Analyze how Credit_Limit affects Personal_loan approval by calculating the average credit limit for clients with and without loans.

LOAN YES

₹

LOAN NO

₹

```
avg_credit_limit_with_loans =
                    CALCULATE(AVERAGE(credit_card[Credit_Limit]),
                    FILTER(customers, customers[Personal_loan] = "yes")
                    )
```

```
avg_credit_limit_without_loan =
                    CALCULATE(AVERAGE(credit_card[Credit_Limit]),
                    FILTER(customers, customers[Personal_loan] = "no")
                    )
```

The 90% watchlist

High Risk Clients Flag: Create a flag for clients whose Total_Revolving_Bal exceeds 90% of their Credit_Limit and who have a high Avg_Utilization_Ratio.

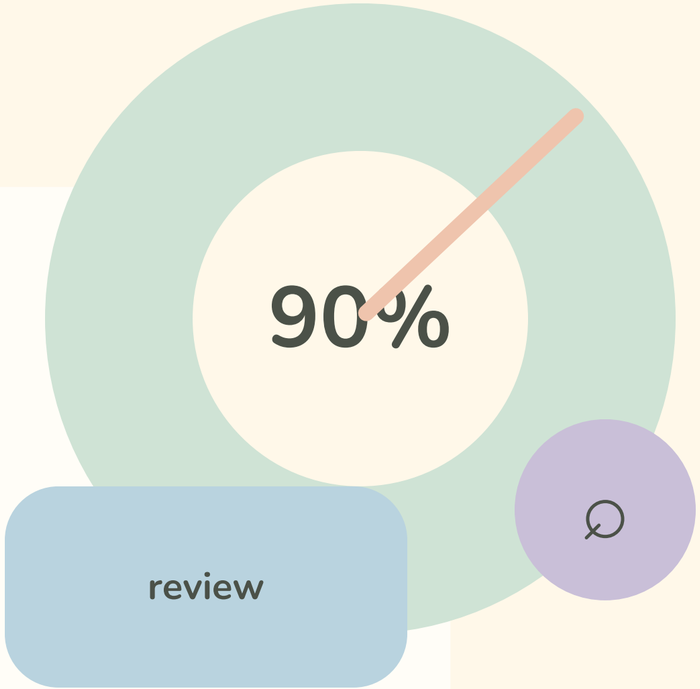
```
risk_flag =

VAR credit_limit = SELECTEDVALUE(credit_card[Credit_Limit])

VAR revolving_bal = SELECTEDVALUE(credit_card[Total_Revolving_Bal])

VAR utilization = SELECTEDVALUE(credit_card[Avg_Utilization_Ratio])

RETURN IF(
            credit_limit > 0
            && revolving_bal > credit_limit * 0.90
            && utilization > 0.80,
           "high_risk", "not_high_risk")
```



Thank You

Data alone is just numbers.
Questions give it direction,
analysis gives it meaning,
and decisions give it value

